Vo Van Tu Tai

Backend Software Engineer (Node.js · NestJS · PHP · PostgreSQL · Redis)

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PROFESSIONAL SUMMARY

Information Systems Engineer (IUH, GPA 3.26/4.0 - Distinction) specializing in scalable backend engineering with **Node.js (NestJS, Express.js) and PHP (OOP, Custom MVC Framework)**. Core technical competencies: 3NF relational database architecture (PostgreSQL, MySQL), B-Tree indexing query tuning (-40% latency), and bulletproof concurrency controls (ACID transactions, pessimistic row locking, and Redis distributed locks). Creator of a distributed logistics OMS & AI tracking platform running in production at **tracking.nexus-ex.site**. Strong capability in reading English API specifications; adept at AI-assisted engineering to drive code quality. Available immediately full-time.

CORE TECHNICAL SKILLS

- **Backend Architecture & APIs:** Node.js (NestJS, Express.js), PHP (OOP, Custom MVC, Clean Architecture), RESTful APIs (OpenAPI 3.0), Webhooks, Idempotency Keys, RBAC, JWT Authentication.
- **Databases & Query Tuning:** PostgreSQL, MySQL (Normalized 3NF, B-Tree Indexes, Composite Indexes, EXPLAIN execution plan optimization), Concurrency Control (ACID Transactions, Pessimistic Row Locking).
- **Message Brokers & In-Memory Cache:** RabbitMQ (Event-Driven Architecture, Decoupled Queue, Fanout/Direct Exchanges), Redis (In-memory Caching, Distributed Locks SETNX with TTL, Geospatial Indexing).
- **DevOps, Testing & Tooling:** Docker, Docker Compose, Linux/Bash basics, Postman API Testing, Git/GitHub (Git Flow, branch management, pull requests), SDLC Lifecycle, Agile/Scrum.
- **AI Tooling & Languages:** Cursor, Claude, ChatGPT, GitHub Copilot (code refactoring & test harness); Technical English (Reading documentation, API specs, and basic communication), Vietnamese (Native).

WORK EXPERIENCE

HomeNest Software
Ho Chi Minh City, Vietnam

Technical Bridge & Backend Engineer (Technical Bridge · Backend / BA Assistant)
Mar 2026 – Sep 2026

Project 1: "Life Assistance" — Multi-Service On-Demand Platform & Dynamic Bidding

- **API Contract Design & Data Modeling:** Modeled relational schemas (ERD 3NF) and OpenAPI 3.0 specifications for an on-demand ecosystem (Handyman, Ride-Hailing, Carpooling, Super Admin); standardized JSON contracts to eliminate client-server schema drift.
- **Dynamic Bidding State-Machine Architecture:** Designed a two-way price negotiation state-machine (customer quote ↔ driver counter-bid) featuring TTL expirations and distributed state locking, preventing race conditions and order deadlocks.
- **Event-Driven GPS Telemetry Optimization:** Proposed event-triggered GPS coordinate logging (strictly upon ride pickup and drop-off milestones rather than continuous polling), cutting server network overhead, reducing disk I/O, and saving third-party map API expenses.
- **Schema Auditing & Release Governance:** Collaborated with Tech Lead to stress-test data concurrency scenarios; audited consistency between PRD documents, database migration scripts, and indexing strategies prior to sprint kickoff.

Project 2: "VietPlus" — Classified Marketplace & Cross-Border Logistics

- **Listing Data Lifecycle Architecture:** Engineered C2C listing lifecycles, automated spam detection filters, and moderation rules; standardized order statuses (Draft → Pending → Active → Expired/Sold) with transactional integrity.
- **Fulfillment, Order Cancellation & Returns State-Machine:** Modeled comprehensive state-machines governing parcel delivery, automated timeout cancellations, and refund dispute escalation workflows.
- **Pragmatic Integration Architecture (Lead Form Intake):** Deployed a lightweight Lead Form module with automated webhook processing for urgent international shipping, successfully safeguarding the sprint's Scope Freeze.
- **API Testing & Defect Triage:** Authored comprehensive Postman API test suites (boundary conditions, negative paths, stress payloads); managed Jira defect triage and facilitated technical UAT sign-offs.

WORK EXPERIENCE (CONTINUED)

J&T Express Ho Chi Minh City, Vietnam

Data Analyst & Operations Coordinator Sep 2025 – Feb 2026

- **Large-Scale SQL Data Reconciliation:** Authored advanced SQL reconciliation queries (Multi-table JOINs, Aggregations, Window Functions) cross-checking physical hub manifests against database event logs; ensured complete data integrity across high-volume shifts.
- **One-Way Operational Throughput Optimization:** Analyzed bottleneck metrics and restructured sorting workflows into a closed one-way sequence (Inbound → Weigh → Scan → Dispatch), clearing hub congestion and shortening shift cycles.
- **Technical SOP Documentation:** Authored operational SOPs standardizing anomaly classification; trained staff in precise system execution, minimizing human data entry errors.

REPRESENTATIVE TECHNICAL PROJECTS

Nexus Express — Distributed Order Management System (OMS) & Live AI Tracking [Live Demo ↗] [GitHub ↗] Distributed Architecture Project

NestJS, TypeScript, PostgreSQL (3NF), RabbitMQ, Idempotency, Live RAG, Docker Compose Sep 2025 – Feb 2026

- **Event-Driven Architecture & RabbitMQ Queues:** Built a production OMS backend live at tracking.nexus-ex.site; utilized NestJS and RabbitMQ to decouple dispatch pipelines from real-time tracking streams, absorbing traffic spikes without blocking main processes.
- **Idempotent API Contracts & Camunda BPMN:** Enforced OpenAPI 3.0 contracts with mandatory Idempotency Keys to prevent duplicate processing during network retries; modeled parcel state lifecycles with Camunda BPMN 2.0.
- **PostgreSQL 3NF Schema & Immutable Audit Logs:** Architected a normalized 3NF schema with composite B-Tree indexes on tracking codes and timestamps; maintained tamper-proof event audit logs enabling 100% transit traceability.
- **Live RAG AI Agent Backend Integration:** Integrated a live RAG assistant via tool/function calling connected directly to backend services, enabling automated parcel tracking and instant rate quotation.

Song Tai Shop & Tourist Accommodation — E-Commerce & Booking Platform [GitHub ↗] Web Application Project

PHP (OOP & Custom MVC), MySQL (3NF & ACID), Webhook VNPay IPN, Concurrency Control, Docker Oct 2025 – Jan 2026

- **Custom PHP MVC Framework & Security:** Developed a custom MVC framework featuring Front-Controller (index.php), REST-style regex routing, PDO prepared statements preventing SQL injections, CSRF tokens, and session-based RBAC.
- **ACID Concurrency Control & TTL Holds:** Prevented inventory overselling and overbooking race conditions using strict MySQL ACID transactions (START TRANSACTION, COMMIT) coupled with pessimistic row locking (SELECT ... FOR UPDATE) and 15-minute temporary holds.
- **Encrypted Payment Webhook Integration:** Integrated VNPay payments using HMAC-SHA512 checksum validation with idempotent webhook handlers; optimized 3NF MySQL schemas with composite indexes achieving sub-second query response times.

Cab Booking & Real-Time Dispatch System — High-Concurrency Backend High-Concurrency System Project

[GitHub ↗]

Node.js, Express.js, Redis (In-Memory Cache & Distributed Locks SETNX), Docker Compose Dec 2025 – Jan 2026

- **High-Concurrency Matching via Redis Distributed Locks:** Built a driver-passenger dispatch engine using Redis distributed locks (SETNX with TTL), eliminating race conditions when multiple drivers simultaneously accept a ride request.
- **In-Memory Caching & Geospatial Queries:** Leveraged Redis geospatial indexes (GEOADD, GEORADIUS) to track real-time driver coordinates and compute proximity in sub-milliseconds, bypassing disk queries on primary databases.
- **Trip State-Machine Governance & Containerization:** Modeled strict ride state lifecycles (Requested → Accepted → Arrived → In-Trip → Completed/Cancelled) with dynamic surge pricing logic; packaged microservices into Docker Compose with health checks.

EDUCATION & ACADEMIC BACKGROUND

Industrial University of Ho Chi Minh City (IUH) Ho Chi Minh City, Vietnam

Engineer in Information Systems (Completed all degree requirements · Available Full-Time) Sep 2022 – 2026

- **Cumulative GPA: 3.26 / 4.0** (Distinction / Xếp loại Giỏi).
- **Relevant Coursework:** Software Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Systems Analysis & Design (OOAD/UML), Advanced Database Systems & SQL (MySQL, PostgreSQL, Indexing, Transactions), SOA, ERP Systems (Odoo).